

Phase equilibria study of Cu–O–ZnO system in various oxygen partial pressures

Longgong Xia^{a,b}, Zhihong Liu^a, Pekka Antero Taskinen^{a,b,*}

^aCentral South University, School of Metallurgy and Environment, Changsha, Hunan 410083, China

^bAalto University, School of Chemical Technology, Metallurgical Thermodynamics and Modeling Research Group, Espoo 02150, Finland

Received 9 September 2015; received in revised form 11 November 2015; accepted 14 December 2015

Available online 23 December 2015

Abstract

Phase relationships of the Cu–O–ZnO system in equilibrium with air ($P_{O_2}=0.21$ atm) and metallic copper (in pure argon) have been studied for the first time using equilibration/quenching/EPMA techniques. In this study, we quantified the phases in equilibrium with air in the temperature range 1095–1500 °C and represented the equilibria by the pseudo-binary phase diagram (Cu₂O–ZnO). The diagram has one simple eutectic reaction with two primary phases of cuprite (Cu₂O) and zincite (ZnO), and the binary eutectic point was found to be 1098 ± 2 °C with 0.10 ± 0.01 mol fraction of ZnO. There were three phases Cu₂O + ZnO + liquid copper or ZnO (Cu₂O) + liquid oxide + liquid copper coexisting with each other in the temperature range 1205–1300 °C, when the system was equilibrated with metallic copper. Isothermal ternary phase diagrams (Cu–O–ZnO) were used to present the results of the experiments at Cu saturation. The eutectic point of the Cu saturated system was determined to be 1208 ± 2 °C and 0.03 ± 0.01 mol fraction of ZnO. Results were compared with the phase diagrams calculated by MTDATA 5.10 and its Mtox 7.0 database. The differences between the experimental data and the computational phase boundaries were significant and the system properties require a re-assessment.

© 2015 Elsevier Ltd and Techna Group S.r.l. All rights reserved.

Keywords: C. Chemical properties; E. Refractories; Advanced ceramics; Microstructure; Smelting

1. Introduction

Zinc oxide is one of the most important and versatile semiconductors with direct wide band gap of 3.37 eV [1]. It has great potential in applications in glass–ceramics [2], solar cells [3], chemical and bio-sensors [4,5]. Copper oxides (both CuO and Cu₂O) are also important functional materials used for catalysis [6], solar-energy transformation [7] and gas sensors [8]. The potential advanced technologies of the Cu–O–ZnO system have been studied extensively during the past decades [9–11].

Reliable thermodynamic descriptions of the Cu–O–ZnO system are significantly important from both academic [12–

17] and industrial perspectives [18–22]. Basic understanding of the phase relationships of this system such as liquidus temperatures contributes a lot in various applications [18–20]. However, experimental data about this system is still incomplete and so are thermodynamic assessments. The only available way of acquiring the phase diagram of this system is by extrapolation, using selected thermodynamics software, such as MTDATA [14], Thermo-Calc [15], Factsage [16], HSC [17]. The extrapolated diagrams must be optimized and validated with real experimental data before being applied reliably in other fields. Possible discrepancies in extrapolations promoted the present experimental study.

Thermodynamics of the Cu–O system have been studied experimentally by numerous independent groups and data compilations, and extensive assessments have been completed [23–32]. Cuprite (Cu₂O) is the stable phase of copper oxide rather than tenorite (CuO) in air at temperatures above 1026 ± 2 °C [23–28]. The melting point of Cu₂O in equilibrium

*Corresponding author at: Aalto University, School of Chemical Technology, Metallurgical Thermodynamics and Modeling Research Group, P.O. Box 16200, Espoo 02150, Finland. Tel.: +358 405017411.

E-mail address: pekka.taskinen@aalto.fi (P.A. Taskinen).

with air has been determined to be approximately 1123 ± 2 °C [23–27]. However, the melting temperature of cuprite (Cu_2O) in equilibrium with metallic copper (Cu) has been reported to be 1223 ± 2 °C [29–32], based on thermodynamic evaluations. Thermodynamic properties of ZnO have been studied extensively [34–39], because of the wide applications of ZnO [1–5,40,41]. The most widely accepted melting point of ZnO is 1975 °C [31,34]. Chapple and Stone [42] studied the ZnO–CuO system in air at 1000 °C, and reported that ZnO solid phase could take up to 10 mol% CuO into solution. Schiavello et al. [43] reinvestigated the Cu–O–ZnO system in the same conditions and stated that the solubility limit in solid ZnO phase was 4 mol% copper. Using X-ray diffraction and electron microscopy, researchers [44–46] reported that ZnO can dissolve 4–8% CuO at 350 °C in air. Unfortunately, no information about the Cu–O–ZnO system at temperatures above 1000 °C has been found in the literature. Solubility of zinc in molten copper at oxide saturation is small [47].

In present study, the phase equilibria of Cu–O–ZnO system in equilibrium with both air ($P_{\text{O}_2}=0.21$ atm) and metallic copper ($P_{\text{O}_2} \approx 10^{-4}$ – 10^{-5} atm) have been investigated using the equilibration/quenching/EPMA technique, which has been proven to be a classic and efficient technique in acquiring high-temperature equilibrium information for geological [48,49] and metallurgical [50] systems. The technique used involved the preparation of a synthetic pellet of predetermined composition. The pellet placed in a suitable container was equilibrated at a target temperature for a certain period of time, followed by rapid quenching, such that the phase assemblages at high-temperatures were frozen instantly. The compositions of all phases were determined by an electron-probe X-ray microanalysis (EPMA) from polished sections. Each specimen

provides information on the liquidus composition and the compositions of the solid phases present in equilibrium. This greatly increases the productivity of phase equilibrium research and offers information e.g. on the solid solution phases, which are of particular importance for further studies, such as thermodynamic modeling [14–17].

2. Material and methods

2.1. Preparation of initial mixtures

High-purity Cu cathode, Cu powder and ZnO powder were employed as starting materials, see Table 1. Cu_2O was prepared by oxidizing the metallic copper cathode in air at 1030 °C for 144 h, and followed by quenching in air. The product was crushed and grinded into fine particles in an agate mortar. According to Hamuyuni [51], the purity of the Cu_2O was more than 99.99%. In the equilibrium experiments, powders were dried at 110 °C, weighed in desired proportions, mixed thoroughly in an agate mortar, and pelletized to 5 MPa in a 5 mm-diameter pressing model. Total weight of the pellet

Table 1

Mass fraction purity and the sources of materials used in the present study.

Chemical	Purity (mass%)	Supplier
Cu cathode	99.996	Boliden Harjavalta (Finland)
Cu powder	99.96	Alfa Aesar (Germany)
ZnO powder	99.99	Alfa Aesar (Germany)
Argon gas	99.999	AGA (Finland)
Platinum wire	99.99	Johnson–Matthey Noble Metals (UK)
Platinum foil	99.99	Johnson–Matthey Noble Metals (UK)

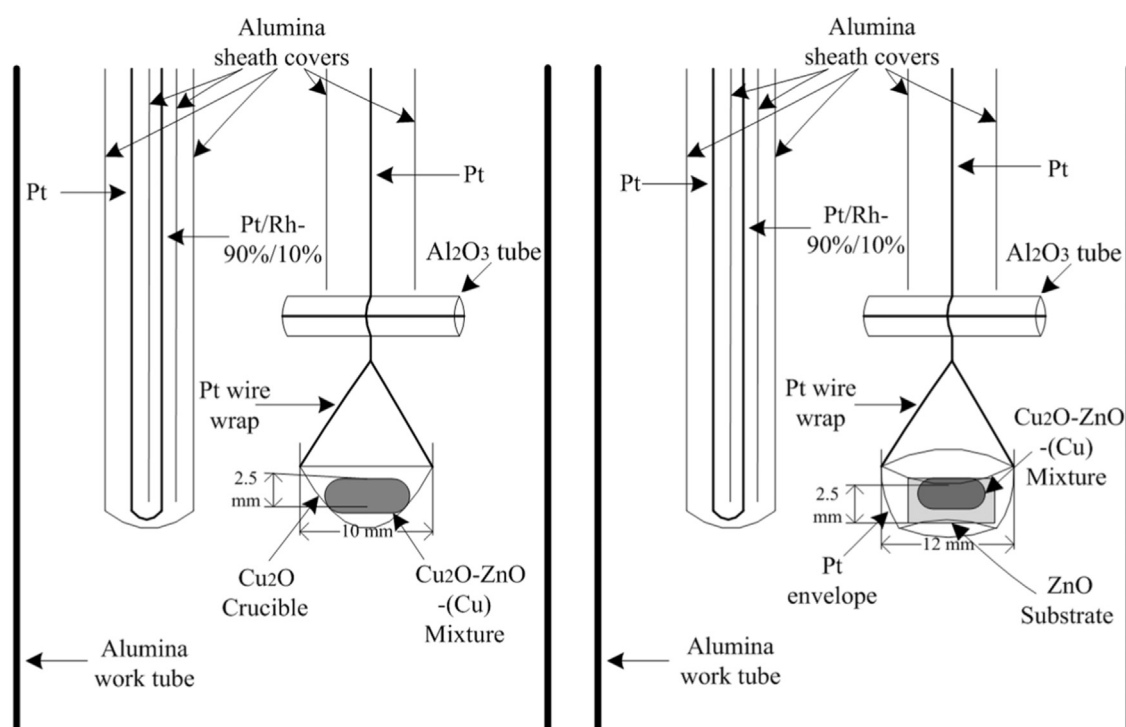


Fig. 1. Schematic of the specimen, container and suspension techniques.

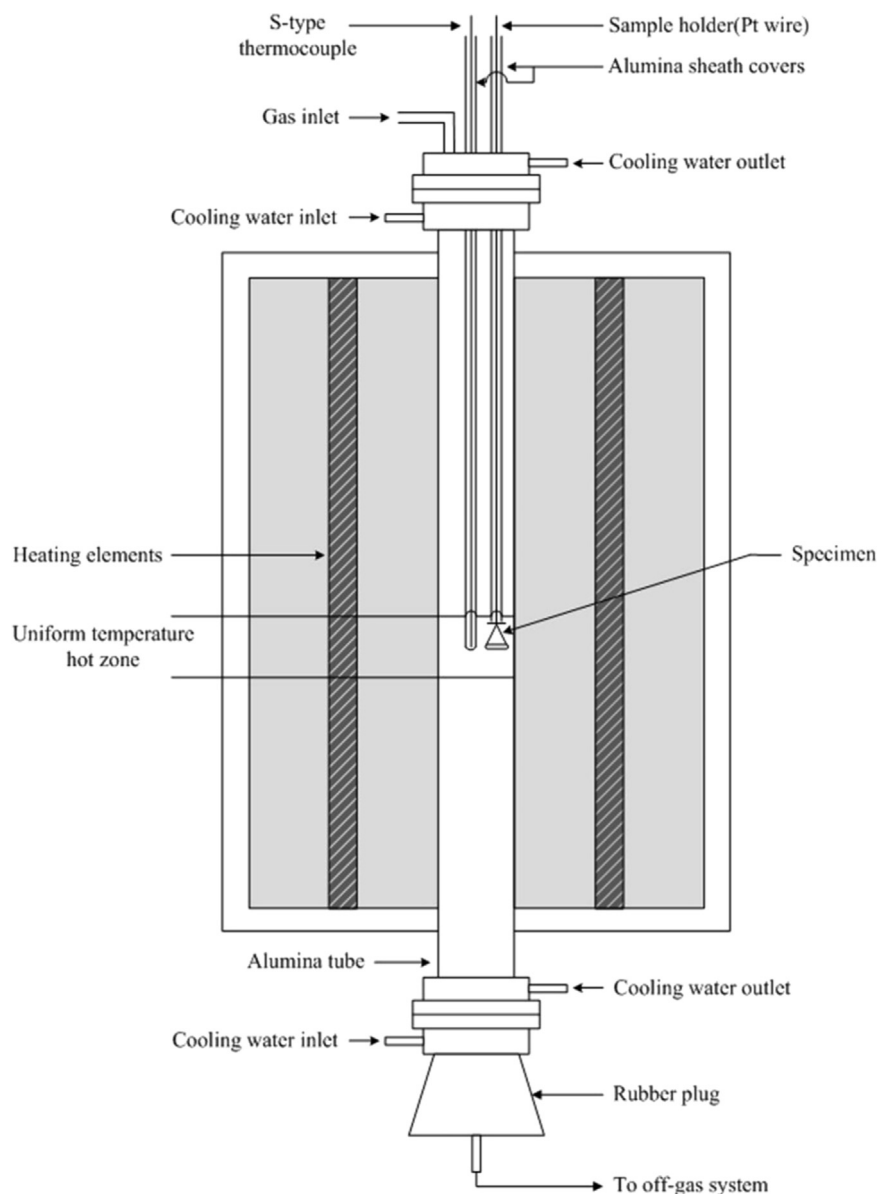


Fig. 2. Schematic vertical section of the furnace and its auxiliaries.

was 0.3 g. The small size of the pellet was an advantage in acquiring equilibria and quenching of molten phases. The pellet was equilibrated at a fixed temperature for a certain period of time, and then quenched in an ice-water bath (0 °C). Pure argon gas-controlled atmosphere was used to conduct the experiment equilibrated with metallic copper, and the purity of the argon gas is presented in Table 1. Platinum wires and foils, which were inert in experimental conditions of this study, were used as suspending and holding wrap to preserve the purity of the system.

2.2. Selection of container material

Experiments on the Cu_2O primary phase field were conducted in self-produced ' Cu_2O crucibles'. A small piece of metallic copper cathode was shaped into a crucible mechanically, oxidizing in air at 1060 °C for 12 h, and then quenched

in air. The composition and purity of the ' Cu_2O crucible' were confirmed by EDS. The 5-mm pellet was placed in the Cu_2O crucible, and equilibrated in a wrap made from 0.5-mm platinum wire (Fig. 1 left). ZnO substrates were employed in experiments in the ZnO saturated domain (Fig. 1 right). The 5-mm pellet was put into a 9 mm-diameter pressing model, and then 0.7 g ZnO powder was added in, pelletized at 10 MPa. The mixture-inlaid pellet was equilibrated in a platinum envelope made from 0.05-mm-thick platinum foil.

2.3. Equilibration technique

The equilibrium experiments were carried out in a vertical tube furnace (LTF 16/-/450, Lenton, UK) with electrical resistance of silicon carbide heating elements (Fig. 2). Special attention was paid to the accuracy of the equilibrium temperature. Uniform temperature hot zone of the furnace was

Table 2

Experimentally determined phase compositions for the Cu–O–ZnO system equilibrated with air ($P_{O_2}=0.21$ atm).*

No.	Temperature (°C) [*]	Time (h)	Phase	EPMA (mol%) [*]		Standard deviation ^{**}
				Cu ₂ O	ZnO	
Cu ₂ O primary-phase field						
1	1120	12	Liquid	91.19	8.81	0.04
			oxide			
			Cu ₂ O	100	0	0.00
ZnO primary-phase field						
2	1095	24	Cu ₂ O	99.37	0.63	0.06
			ZnO	0.48	99.52	0.07
3	1100	12	Liquid	89.50	10.50	0.13
			oxide			
			ZnO	2.54	97.46	0.11
4	1150	12	Liquid	88.83	11.17	0.16
			oxide			
			ZnO	1.55	98.45	0.08
5	1200	4	Liquid	88.74	11.26	0.16
			oxide			
			ZnO	1.76	98.24	0.14
6	1250	4	Liquid	87.76	12.24	0.09
			oxide			
			ZnO	1.36	98.64	0.13
7	1300	4	Liquid	86.86	13.14	0.08
			oxide			
			ZnO	1.71	98.29	0.11
8	1350	4	Liquid	84.21	15.79	0.22
			oxide			
			ZnO	1.42	98.58	0.17
9	1400	4	Liquid	83.12	16.88	0.17
			oxide			
			ZnO	1.32	98.68	0.11
10	1450	4	Liquid	82.90	17.10	0.26
			oxide			
			ZnO	1.42	98.58	0.11
11	1500	0.5	Liquid	81.19	18.11	0.05
			oxide			
			ZnO	0.99	99.01	0.07

*Standard uncertainty: $\mu(T)=\pm 1$ °C; $\mu(P_{O_2})=\pm 5\%$; $\mu(X)=\pm 0.01\%$.

**Average standard deviation: $\sigma(\text{EPMA})=0.09$.

confirmed by measuring its thermal profiles at several temperatures from 1000 °C to 1600 °C. The temperature of the hot zone was determined with a calibrated S-type thermocouple (Johnson–Matthey Noble Metals, UK), which was connected to a Keithley 2010 DMM multimeter. The S-type thermocouple was calibrated with metallic copper, zinc and pure water. The collected phase transition temperatures (solid $\leftrightarrow$ liquid) of copper, zinc and water were 1083.61 ± 0.2 °C, 419.25 ± 0.1 °C and 0.17 ± 0.1 °C, respectively, which were consistent with literature data [13,15,19]. The working thermocouple in a recrystallized alumina sheath was positioned immediately next to the specimen and periodically tested against a calibrated thermocouple. The cold junction compensation was performed by a Pt100 resistance thermometer (SKS-group, Finland, tolerance class B 1/10), connected to a Keithley 2000 DMM multimeter. The overall temperature accuracy was estimated to be ± 1 °C for the entire experimental temperature range of interest. The temperature data

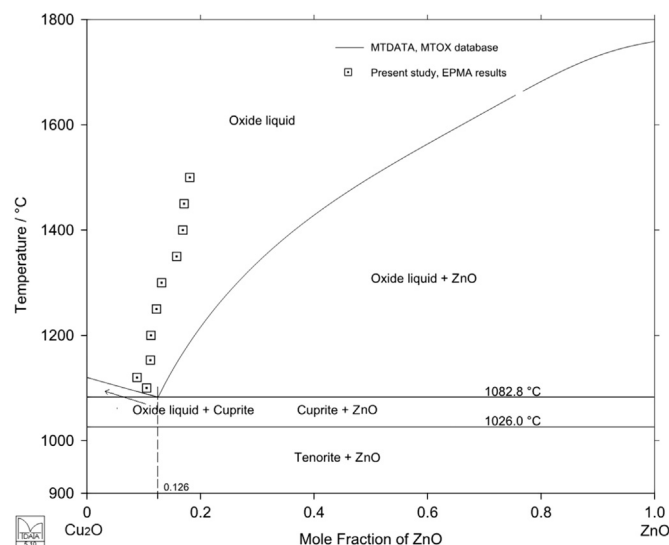


Fig. 3. Isoleth of a quasi-binary section of the Cu–O–ZnO system in air ($P_{O_2}=0.21$ atm) (Cu₂O–ZnO) with the experimental points (□) superimposed.

during the experiments were collected with a NI LabVIEW temperature logging program.

The specimen hooked to a pure platinum wire (0.5 mm diameter) was introduced into the furnace from the bottom of the working tube and was suspended in the hot zone. The specimen was kept in the hot zone, at the target temperature, for the required equilibration period. Particular attention was paid to ensure the achievement of equilibria, by approaching the final equilibrium compositions with variation of time. A longer equilibrium time is a positive factor to approaching the equilibrium. Achievement of the equilibrium was determined by measuring the composition of the liquid phase from different times. Three dedicated series of experiments at different temperatures (1200, 1350, 1500 °C) were conducted to determine the required equilibrium period. The results indicated that homogenous phases could be achieved within 4 h at 1200 °C, and the proper equilibration period at 1350 °C and 1500 °C was 2 h and 0.25 h, respectively.

The experiments equilibrated with air were conducted with the bottom and the top of the reaction tube open to the ambient atmosphere in order to ensure the composition of air. For metallic copper saturated experiments, an inert purified argon gas was continuously flushed through the gas tight system. The specimen was introduced into the working tube by a platinum wire, and hanged at the bottom. The working tube was sealed with a rubber plug after the specimen was introduced in. The specimen was lifted up to the hot zone after 30 min, because the specimen was easily oxidized by the possible residual oxygen in the working tube. Once, the equilibration time was reached, the bottom of the tube was immersed into a bottle of water–ice mixture. The specimen was then released and quenched rapidly into an ice–water bath (0 °C). Afterwards, it was dried and mounted in an epoxy resin. A polished cross section of the mounted specimen was prepared using conventional metallographic grinding and polishing techniques.

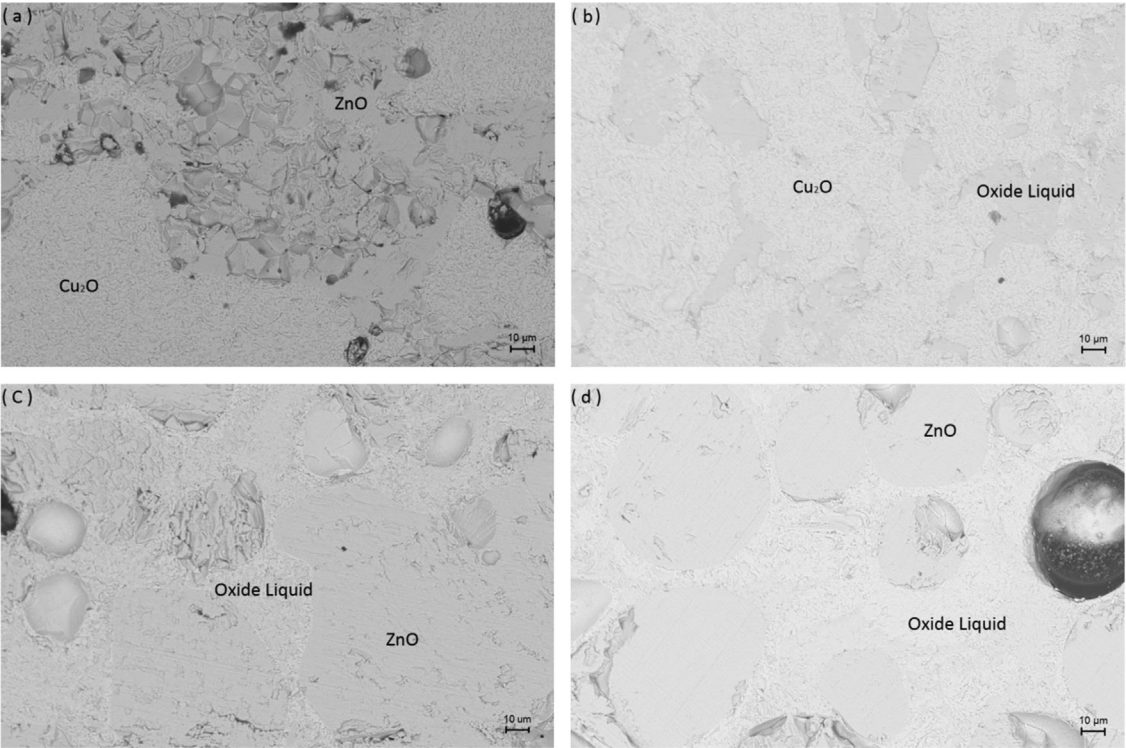


Fig. 4. Backscattered SEM images of quenched Cu–O–ZnO samples equilibrated with air: (a) at 1095 °C for 24 h; (b) at 1120 °C for 12 h; (c) at 1100 °C for 12 h; and (d) at 1400 °C for 4 h.

Table 3
Experimentally determined phase compositions for the Cu–O–ZnO system equilibrated with metallic copper (in argon atmosphere).

No.	Temperature (°C) [*]	Time (h)	Phase	EPMA (mol%) [*]			Standard deviation ^{**}
				Cu	ZnO	O	
Triple-point of liquid Cu, oxide liquid and Cu ₂ O							
1	1220	12	Liquid	63.33	1.76	34.91	0.15
			oxide				
			Cu ₂ O	66.63	60.01	33.36	0.17
Triple-point of liquid Cu, oxide liquid and ZnO							
2	1205	24	Cu ₂ O	66.03	0.17	33.80	0.03
			ZnO	0.99	90.91	8.10	0.12
3	1210	12	Liquid	63.55	33.42	33.03	0.17
			oxide				
			ZnO	1.03	89.78	9.19	0.18
4	1250	4	Liquid	61.83	4.37	33.80	0.16
			oxide				
			ZnO	1.13	87.60	11.27	0.09
5	1300	4	Liquid	61.25	4.62	34.13	0.18
			oxide				
			ZnO	1.30	86.72	11.98	0.10

*Standard uncertainty: $\mu(T)=\pm 1\text{ }^{\circ}\text{C}$; $\mu(X)=\pm 0.01\%$.
** Average standard deviation: $\sigma(\text{EPMA})=0.12$.

2.4. Analysis

All samples were carbon-coated with a Leica EM SCD050 Coater (supplied by Leica Mikrosysteme GmbH, Vienna, Austria). The microstructure was examined by a LEO 1450 (Carl Zeiss Microscopy GmbH, Jena, Germany) scanning

electron microscope. Compositions of the equilibrated phases were then measured by Electron-Probe X-ray Microanalysis (EPMA). The analysis was performed by a CAMECA SX100 instrument (Cameca, France), which was equipped with five wavelength dispersive spectrometers. An accelerating voltage of 15 kV, beam current of 40 nA, and spot size of 1–100 μm were utilized in the analysis. The standards used in this work were natural minerals (sphalerite for Zn: Kα metallic copper for Cu: Kα hematite for O: Kα). All standards were purchased from Astimex Scientific Ltd., Canada. The raw measurement results were corrected using the PAP on-line matrix correction program [52].

3. Results and discussion

3.1. The Cu–O–ZnO system equilibrated at $P_{O_2}=0.21\text{ atm}$

Basing on SEM and EPMA analysis, each samples contained two homogeneous phases, and no concentration gradients were detected within them. Chemical compositions of those phases in equilibrium were carefully measured with EPMA. At least 10 independent measurement points from each phase were analyzed to quantify its accuracy, and those points came from different, well quenched parts of the sample. The standard uncertainty of each measurement point in EPMA analyzing is 0.01 mol%. The obtained EPMA results of the Cu–O–ZnO system in equilibrium with air at temperatures range from 1095 °C to 1500 °C are listed in Table 2. The standard deviation of each phase has been calculated and

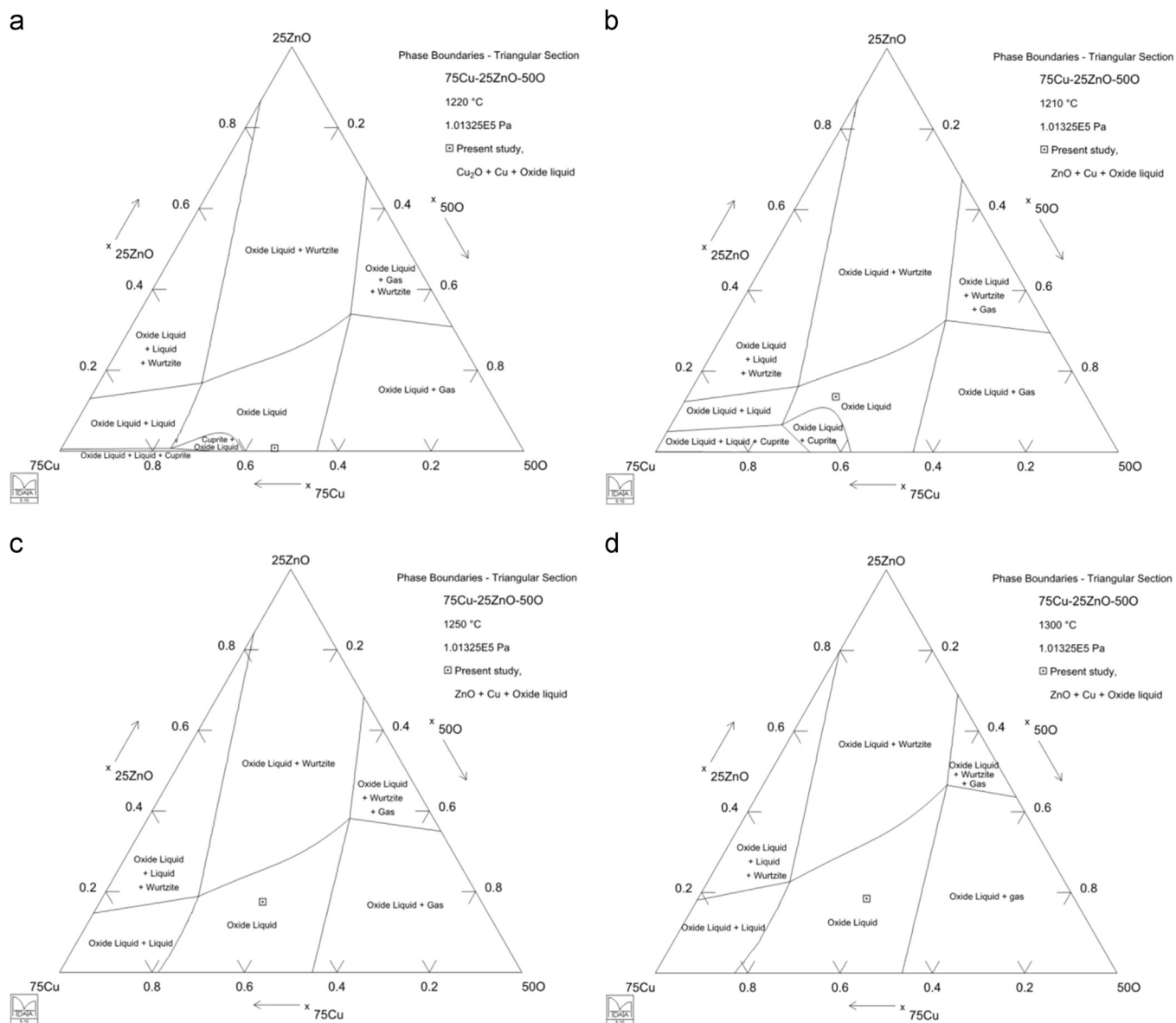


Fig. 5. Calculated Gibbs triangles of the Cu–O–ZnO ternary system with 75 at% Cu, 50 at% O and 25 at% ZnO as the terminal phases, equilibrated at: (a) 1220 °C; (b) 1210 °C; (c) 1250 °C; and (d) 1300 °C, with the experimental points (□) superimposed.

presented in the Table 2. Overall standard deviation of EPMA of all experimental points is also given below the table.

For the temperature range of interest, the concentration of ZnO of the liquid phase at the ZnO liquidus shows an increasing tendency along with increasing temperature. The Cu_2O solubility in the terminal solid phase of ZnO was 2.54 mol% at 1100 °C, and decreased to 0.99 mol% at 1500 °C. No solubility of ZnO in the solid Cu_2O terminal phase was observed, according to the current experimental work, see Table 2.

The results of present study have been compared with the assessed phase diagram computed by MTDATA (version 5.10) [53] using MTOX 7.0 database [54] and a comparison is depicted graphically in Fig. 3. According to the modeling, there is no mixed CuO_x –ZnO compound formed in this system, and the binary eutectic is located at 1082.8 °C at 0.126 mol

fraction of ZnO. Tenorite is transformed into cuprite at 1026.0 °C in air, which is in agreement with previous studies [55,56]. The concentration of ZnO in the liquid oxide equilibrated at 1120 °C in the Cu_2O primary phase field is according to these observations $x_{\text{ZnO}}=0.088$, which is higher than the calculated value $x_{\text{ZnO}}=0.023$. However, the experimental concentration of ZnO in the liquid oxide phase at ZnO primary phase field is much lower than the calculated one.

The eutectic reaction between cuprite, zincite and oxide liquid was found in this study to occur at 1098 ± 2 °C and 0.10 ± 0.01 mol fraction of ZnO. As shown in Fig. 4(a), two solid phases (Cu_2O and ZnO) were detected in the sample equilibrated at 1095 °C. In the similar way, the sample equilibrated at 1100 °C at the ZnO primary phase field, contains one liquid phase and one solid phase, as shown in Fig. 4(c). It is

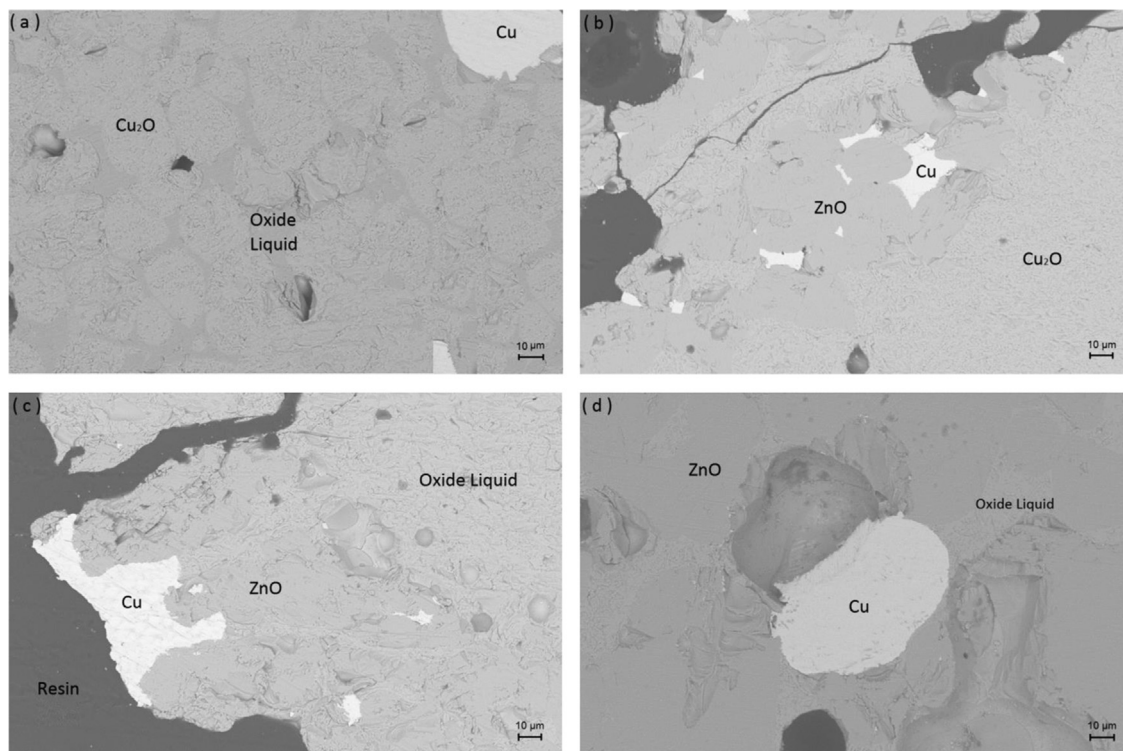


Fig. 6. Backscattered electron (SEM) images of quenched Cu–O–ZnO samples equilibrated with metallic copper: (a) at 1220 °C for 12 h; (b) at 1205 °C for 24 h; (c) at 1210 °C for 12 h; and (d) at 1300 °C for 4 h.

obvious that the area of liquid grows gradually wider along with increasing temperature, according to Fig. 4(c) and (d).

3.2. The Cu–O–ZnO system equilibrated with metallic copper

The critical point of the Cu–O binary system and its liquid phase immiscibility region is 1347 ± 3 °C, based on the independent studies of Clavaguera-Mora [25], Taskinen [33] and Rannikko [57]. The melting point of cuprite (Cu_2O) in equilibrium with metallic copper (Cu) was reported to be 1223 ± 2 °C [29–32], based on thermodynamic evaluations.

Five experiments have been conducted in the temperature range of 1205–1300 °C to clarify and approach the equilibria of the Cu–O–ZnO ternary system in equilibrium with metallic copper. Selected pseudo-ternary phase diagrams (Cu–O–ZnO) have been used to present the results as isothermal sections throughout the different phase fields. The obtained EPMA results in equilibrium with metallic copper are listed numerically in Table 3. The standard deviations of the measurements were calculated from the EPMA data and the values are presented in Table 3.

For the temperature range of interest, the solubility of Cu in the solid phase of ZnO ranges from 0.99 mol% to 1.30 mol%. The concentration of ZnO in the liquid oxide phase increases gradually along with temperature. The atomic ratio of copper to oxygen in the liquid oxide phase here is higher than 2, because the phase boundary saturated with metallic copper locates on more copper-rich compositions than Cu_2O . The results of the present study have been compared with the isothermal sections calculated by MTDATA (version 5.10)

[53] with MTOX 7.0 database [54] and the computational phase boundaries with the experimental points are depicted graphically in Fig. 5.

For better comparison, an enlargement of the Cu_2O -rich region has been presented in Fig. 5. According to the 75Cu–25ZnO–50O triangular section equilibrated at 1210 °C in Fig. 5 (b), there are three triple-points in this system: the first one with metallic copper, liquid oxide and cuprite co-existing; the second one with metallic copper, liquid oxide and zincite coexisting; the third one with liquid oxide, zincite and gas coexisting. Fig. 5 (a) shows the result of experiment no.1 (Table 3), which has metallic copper, liquid oxide and cuprite present and in equilibrium with each other. Above the melting point of cuprite (Cu_2O , 1223 ± 2 °C), the first triple-point disappears, showing in the triangles calculated at 1250 °C (Fig. 5(c)) and 1300 °C (Fig. 5(d)). With increasing temperature, the area of liquid oxide grows wider, which is important and beneficial for high-temperature processes [19,20]. The concentrations of ZnO in the experimental points are relatively close to the calculated ones. However, the calculated concentrations of Cu in the molten oxide phase are about 20% higher than the current experimental values.

The microstructure of the specimen equilibrated in the Cu_2O primary phase field (Fig. 6(a)) is quite different from the ones in the ZnO primary phase field; the ZnO crystals are easier to distinguish by their shape. The ternary eutectic reaction was found to occur at 1208 ± 2 °C and 0.03 ± 0.01 mol fraction of ZnO in this study. As shown in Fig. 6(b), two solid phases (Cu_2O and ZnO) and one liquid phase (metallic Cu) are detected in the sample equilibrated at 1205 °C. In a similar

way, the specimen equilibrated at 1210 °C contained two liquid phases (metallic copper and liquid oxide) and one solid phase (ZnO), which are shown in Fig. 6(c). Temperature has a positive impact for infiltrating the solid phases by the melts, according to Fig. 6(c) and (d).

4. Conclusions

The equilibration/quenching/EPMA analysis technique has been successfully employed to determine the phase equilibria and liquidus temperatures of the Cu–O–ZnO system in equilibrium with air, and with metallic copper. No ternary compound has been detected in the experimental samples. Concentrations of ZnO and copper along the liquidus lines of the Cu₂O and ZnO primary phase fields of the system have been measured at 1095 to 1500 °C.

When equilibrated with air, one simple eutectic reaction with two primary phase fields of cuprite (Cu₂O) and zincite (ZnO), respectively, has been observed. The binary eutectic point with this constraint was found to be 1098 ± 2 °C with 0.10 ± 0.01 mol fraction of ZnO in the liquid oxide phase. Also the solubility of Cu₂O in ZnO terminal phase was determined.

The phase equilibria of the Cu–O–ZnO system (1205–1300 °C) changed significantly, when the slag was equilibrated with metallic copper. The concentration of ZnO in the liquid oxide phase increased only gradually along with temperature. The eutectic point was determined to be at 1208 ± 2 °C with 0.03 ± 0.01 mol fraction of ZnO. The obtained liquidus lines have been compared with the extrapolated isothermal phase diagrams, calculated using MTDATA 5.10 software, and the differences with the experimental data are significant at temperatures above 1200 °C.

It should be pointed out that the experimental data provided in the present study represent the first experimental determination of the liquidus temperatures of the Cu–O–ZnO system and, as such, should provide novel data for both scientific research and industrial practices.

Acknowledgments

The present study is a part of a joint program between Central South University and Aalto University, and the work was conducted in Aalto University. Great thanks is to be expressed to Mr. Joseph Hamuyuni for the valuable discussions.

The research was financially supported by Association of Finnish Steel and Metal Producers and Systems Integrated Metal Processes (SIMP) program by FIMECC and Tekes. The authors also acknowledge Electron Microprobe Laboratory of Geological Survey of Finland for the EPMA analysis.

References

- [1] U. Ozgur, Y.I. Alivov, C. Liu, A. Teke, M.A. Reshchikov, S. Dogan, V. Avrutin, S.J. Cho, H. Morkoc, A comprehensive review of ZnO materials and devices, *J. Appl. Phys.* 98 (2005) 041301.
- [2] J. Zou, Q. Zhang, H. Yang, H. Sun, A new system of low temperature sintering ZnO–SiO₂ dielectric ceramics, *Jpn. J. Appl. Phys.* 45 (5) (2006) 4143–4145.
- [3] R. Seema, S. Poonam, P.K. Shishodia, R.M. Mehra, Synthesis of nanocrystalline ZnO powder via sol–gel route for dye-sensitized solar cells, *Sol. Energy Mater. Sol. Cells* 92 (12) (2008) 1639–1645.
- [4] R. Kumar, O. Al-Dossary, G. Kumar, A. Umar, Zinc oxide nanostructures for NO₂ gas-sensor applications: a review, *Nano-Micro Lett.* 7 (2) (2015) 97–120.
- [5] S. Arya, S. Saha, J.E. Ramirez-Vick, V. Gupta, S. Bhansali, S.P. Singh, Recent advances in ZnO nanostructures and thin films for biosensor applications: review, *Anal. Chim. Acta* 737 (6) (2012) 1–21.
- [6] J. Gong, H. Yue, Y. Zhao, S. Zhao, L. Zhao, J. Lv, S. Wang, X. Ma, Synthesis of ethanol via syngas on Cu/SiO₂ catalysts with balanced Cu⁰–Cu⁺ Sites, *J. Am. Chem. Soc.* 134 (34) (2012) 13922–13925.
- [7] C. Schindler, M. Weides, M.N. Kozicki, R. Waser, Low current resistive switching in Cu–SiO₂ cells, *Appl. Phys. Lett.* 92 (2008) 122910.
- [8] J. Zhang, J. Liu, Q. Peng, X. Wang, Y. Li, Nearly monodisperse Cu₂O and CuO nanospheres: preparation and applications for sensitive gas sensors, *Chem. Mater.* 18 (4) (2006) 867–871.
- [9] I.T. Drapak, Visible luminescence of a ZnO–Cu₂O heterojunction, *Semiconductor* 2 (1968) 624.
- [10] H. Ohta, K. Kawamura, M. Orita, N. Sarukura, M. Hirano, H. Hosono, Current injection emission from a transparent p–n junction composed of p-SrCu₂O₂/n-ZnO, *Appl. Phys. Lett.* 77 (2000) 475–477.
- [11] Y. Chen, C. Liao, Y. Chueh, C. Lai, L. Chen, A. Chu, C. Kuo, H. Wang, High performance Cu₂O/ZnO core-shell nanorod arrays synthesized using a nanoimprint GaN template by the hydrothermal growth technique, *Opt. Mater. Express* 4 (7) (2014) 1473–1486.
- [12] H. Lukas, S.G. Fries, B. Sundman, *Computational Thermodynamics: The Calphad Method* ISBN, Cambridge University Press, New York (USA), ISBN 978-0-521-86811-2, 2007.
- [13] A.T. Dinsdale, SGTE data for pure elements, *Calphad* 15 (4) (1991) 317–425.
- [14] R.H. Davies, A.T. Dinsdale, T.G. Chart, T.I. Barry, M.H. Rand, Application of MTDATA to the Modeling of Multicomponent Equilibria, *High Temperature Science*, 1, , 1990, p. 251–262.
- [15] B. Sundman, B. Jansson, J.O. Andersson, The Thermo-calc databank system, *Calphad* 9 (2) (1985) 153–190.
- [16] C.W. Bale, P. Chartrand, S.A. Degterov, G. Eriksson, K. Hack, R. Ben Mahfoud, J. Melancon, A.D. Pelton, S. Petersen, FactSage thermochemical software and databases, *Calphad* 26 (2) (2002) 189–228.
- [17] A. Roine, HSC Chemistry, Outokumpu Research Oy, Pori, Finland. (www.hsc-chemistry.com), 2002.
- [18] T.G. Nieh, J. Wadsworth, O.D. Sherby, *Superplasticity in Metals and Ceramics* ISBN, Cambridge University Press, New York, USA, ISBN 0-521-56105-1, 1997.
- [19] F. Habashi, in: *Handbook of Extractive Metallurgy*, Wiley-Vch, Heidelberg, Germany, 1997.
- [20] M.E. Schlesinger, M.J. King, K.C. Sole, W.G. Davenport, *Extractive Metallurgy of Copper*, fifth edition, Elsevier Science Ltd., Oxford, UK, 2011 ISBN 978-0-08-096789-9.
- [21] T.J. Hsueh, C.L. Hsu, S.J. Chang, P.W. Guo, J.H. Hsieh, I.C. Chen, Cu₂O/n-ZnO nanowire solar cells on ZnO: Ga/glass templates, *Scr. Mater.* 57 (1) (2007) 53–56.
- [22] J. Katayama, K. Ito, M. Matsuoka, J. Tamaki, Performance of Cu₂O/ZnO solar cell prepared by two step electrodeposition, *J. Appl. Electrochem.* 34 (2004) 687–692.
- [23] L. Schramm, G. Behr, W. Loser, K. Wetzig, Thermodynamic reassessment of the Cu–O phase diagram, *J. Phase Equilibria Diffus.* 26 (6) (2005) 605–612.
- [24] D. Shishin, S.A. Decterov, Critical assessment and thermodynamic modeling of the Cu–O and Cu–O–S systems, *Calphad* 38 (1) (2012) 59–70.
- [25] M.T. Clavaguera-Mora, J.L. Tournon, J. Rodriguez-Viejo, N. Clavaguera, Thermodynamic description of the Cu–O system, *J. Alloys Compd.* 377 (1–2) (2004) 8–16.

- [26] B. Hallstedt, D. Risold, L.J. Gauckler, Thermodynamic assessment of the copper–oxygen system, *J. Phase Equilib.* 15 (5) (1994) 483–499.
- [27] R. Schmid, A thermodynamic analysis of the Cu–O system with an associated solution model, *Metall. Trans. B* 14 (3) (1983) 473–481.
- [28] K.W. Lay, G.M. Renlund, Oxygen pressure effect on the Y_2O_3 –BaO–CuO Liquidus, *J. Am. Ceram. Soc.* 73 (5) (1990) 1208–1213.
- [29] R.D. Holmes, H.S. O'Neill, R.J. Arculus, Standard gibbs free energy of formation for Cu_2O , NiO, CoO and Fe_3O_4 : high resolution electrochemical measurements using zirconia solid electrolytes from 900–1400 K, *Geochim. Cosmochim. Acta* 50 (11) (1986) 2439–2452.
- [30] T. Peters, Equilibrium fugacities of the Cu_2O –CuO and Cu–CuO buffers, *Schweiz. Mineral. Petrogr. Mitt* 63 (1983) 7–11.
- [31] M.W. Chase, J.L. Curnutt, J.R. Downey, R.A. McDonald, A.N. Syverud, E.A. Valenzuela, JANAF thermodynamical tables, 1982 supplement, *J. Phys. Chem. Ref. Data* 11 (1982) 695–940.
- [32] A. Boudene, K. Hack, A. Mohammad, D. Neuschuetz, E. Zimmermann, Experimental investigation and thermodynamical assessment of the system Cu–O, *Z. Metallkunde* 24 (9) (1992) 663–668 ISSN 0044-3093.
- [33] P. Taskinen, The standard gibbs energy of formation of Cu_2O (s) at 1066–1220 °C, *Scand. J. Metall.* 10 (4) (1981) 189–191.
- [34] L.B. Pankratz, R.V. Mrazek, Thermodynamic properties of elements and oxides, *US. Bur. Mines Bull.* 672 (1982) 509.
- [35] I. Satoshi, A. Takeshi, Thermodynamic studies of ZnO in solid zinc silicate: fundamental studies of zinc extraction by the iron-reduction distillation process, *J. MMIJ* 105 (9) (1989) 685–692 4th Report.
- [36] H.E. Brown, Zinc Oxide: Properties and Applications, International Lead Zinc Research Organization, New York, 1975.
- [37] E.C. Truesdale, R.K. Waring, Reduction equilibria of zinc oxide and carbon monoxide, *J. Am. Chem. Soc.* 63 (6) (1941) 1610–1621.
- [38] C. Ma, Q. Shi, B.F. Woodfield, A. Navrotsky, Low temperature heat capacity of bulk and nanophase ZnO and $Zn_{1-x}Co_xO$ wurtzite phases, *J. Chem. Thermodyn.* 60 (2013) 191–196.
- [39] K. Kusaba, Y. Syono, T. Kikegawa, Phase transition of ZnO under high pressure and temperature, *Proc. Jpn. Acad. Ser. B: Phys. Biol. Sci* 75 (1999) 1–6.
- [40] E. Fortunato, P. Barquinha, A. Pimentel, A. Goncalves, A. Marques, L. Pereira, R. Martins, Recent advances in ZnO transparent thin film transistors, *Thin Solid Films* 487 (1) (2005) 205–211.
- [41] M. Saito, S. Fujihara, Large photocurrent generation in dye-sensitized ZnO solar cells, *Energy Environ. Sci.* 1 (2008) 280–283.
- [42] F.H. Chapple, F.S. Stone, The optical properties of the cupric ion in oxide crystal fields of differing symmetries, *Proc. Br. Ceram. Soc.* 1 (1964) 45–58.
- [43] M. Schiavello, F. Pepe, S.D. Rossi, The catalytic activity of copper ions in copper oxides and in their solid solutions with ZnO and MgO for the decomposition of N_2O , *Phys. Chem.* 92 (8) (1974) 109–124.
- [44] R.G. Herman, K. Klier, G.W. Simmons, B.P. Finn, J.B. Bulko, Catalytic synthesis of methanol from CO/H_2 : I. Phase composition, electronic properties, and activities of the Cu/ZnO/ M_2O_3 catalysts, *J. Catal.* 56 (3) (1979) 407–429.
- [45] S. Mehta, G.W. Simmons, K. Klier, R.G. Herman, Catalytic synthesis of methanol from CO/H_2 : II. Electron microscopy (TEM, STEM, micro-diffraction, and energy dispersive analysis) of the Cu/ZnO and Cu/ZnO/ Cr_2O_3 catalysts, *J. Catal.* 57 (3) (1979) 339–360.
- [46] J.B. Bulko, R.G. Herman, K. Klier, G.W. Simmons, Optical properties and electronic interactions of microcrystalline copper/zinc oxide (Cu/ZnO) catalysts, *J. Phys. Chem.* 83 (24) (1979) 3118–3122.
- [47] S. Seetharaman, L.-I. Staffansson, Thermodynamic Studies of Zn–O interactions in dilute liquid copper alloys, *Metall. Trans. B* 10 (4) (1979) 539–543.
- [48] M.W. Grutzeck, A. Muan, Phase relations at liquidus temperatures in the system MgO–NiO– SiO_2 , *J. Am. Ceram. Soc.* 71 (8) (1988) 638–643.
- [49] P. Hudon, I. Jung, D.R. Baker, Experimental investigation and optimization of thermodynamic properties and phase diagrams in the systems CaO– SiO_2 , MgO– SiO_2 , $CaMgSi_2O_6$ – SiO_2 and $CaMgSi_2O_6$ –Mg $_2SiO_4$ to 1.0 GPa, *J. Petrol.* 46 (9) (2005) 1859–1880.
- [50] E. Jak, P.C. Hayes, Phase equilibria determination in complex slag systems, *Trans. Inst. Mining Metall. C* 117 (1) (2008) 1–17.
- [51] J. Hamuyuni, N. Hellsten, G. Akdogan, P. Taskinen, The liquidus in Cu–O–CaO system at metallic copper saturation up to 1698 K, *J. Am. Ceram. Soc.* 98 (1) (2015) 320–323.
- [52] J.L. Pouchou, F. Pichoir, Basic Expression of "PAP" Computation for Quantitative EPMA, in: J.D. Brown, R.H. Packwood (Eds.), *Proceedings of the 11th International Congress on X-ray Optics and Microanalysis (ICXOM)*, Ontario, Canada. ICXOM-11, Ontario, Canada, 1987.
- [53] MTDATA ver. 5.10.2011, NPL, Teddington, U.K. (www.npl.co.uk/science-technology/mathematics-modelling-and-simulation/mtdata/), 2011.
- [54] MTOX, Release Notes for Version 7.0 of Mtox Database, National Physical Laboratory, Teddington, UK, 2011.
- [55] Z.B. Shao, K.R. Liu, L.Q. Liu, H.K. Liu, S.X. Dou, Equilibrium phase diagram in the system PbO–Ag and CuO–Ag, *J. Am. Ceram. Soc.* 76 (10) (1993) 2663–2664.
- [56] H. Nishiura, R.O. Suzuki, K. Ono, L.J. Gauckler, Experimental phase diagram in the Ag– Cu_2O –CuO System, *J. Am. Ceram. Soc.* 81 (8) (1998) 2181–2187.
- [57] H. Rannikko, A thermodynamic assessment of the Cu–O–CaO– SiO_2 system, *Acta Polytech. Scand. Chem. Technol. Ser.* 229 (1995) 1–78.